Seungeun Lee

Professional Research Personnel, Republic of Korea

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RESEARCH INTEREST

Machine Learning, Computer Vision, and Medical Imaging, Including

- Geometric Deep Learning on Sphere
  - Spherical Harmonic Analysis
  - Generative Modeling on Sphere
- Medical Imaging
  - Cortical Surface Registration
  - Sulcal Labeling
- 3D Human Modeling
  - 3D avatar reconstruction and animation

EDUCATION

M.S., Computer Science and Engineering, UNIST 2021–2023
Advisor: Prof. Ilwoo Lyu

B.S., Information & Communication Engineering, Inha University 2017-2021
Summa Cum Laude (rank: 3/89), GPA: 4.2/4.5, Total credit: 152

WORK EXPERIENCE

Professional Research Personnel, Republic of Korea Nov. 2023–Present
Advisor: Prof. Gyeong-Moon Park

- Researched and developed models for dynamic appearance reconstruction and animation of full-body 3D avatars wearing loose clothing such as skirts.
- Researched and developed models for audio-driven talking avatar reconstruction and animation, including facial expressions and gestures.

Research Intern, NAVER CLOVA Jul. 2022–Oct. 2022

- Developed a human motion capture model that estimates robust 3D human poses for a healthcare system providing posture correction services.

SELECTED PUBLICATIONS (1ST AUTHOR)

[C4] **Seungeun Lee**, SeungJun Moon, Hah Min Lew, Ji-Su Kang, Gyeong-Moon Park, "Personalized Audio-driven Whole-body Talking Avatars," *CVPR*, 2026.

[C3] **Seungeun Lee**, SeungJun Moon, Hah Min Lew, Ji-Su Kang, Gyeong-Moon Park, "Secondary Motion-aware 3D Gaussian Avatars for Modeling Dynamic Appearances," *ICLR*, 2026.

[C2] **Seungeun Lee**, Sergey Pyatkovskiy, Jaejun Yoo, Ilwoo Lyu, "Spherical Diffusion Process for Score-Guided Cortical Correspondence via Spectral Attention," *MICCAI*, 2025.

[J2] **Seungeun Lee**, Seunghwan Lee, Ethan Willbrand, Benjamin Parker, Silvia Bunge, Kevin Weiner, Ilwoo Lyu, "Leveraging Input-Level Feature Deformation with Guided-Attention for Sulcal Labeling," *IEEE Transcation on Medical Imaging*, 2024.

[J1] **Seungeun Lee**, Seunghwan Lee, Sunghwa Ryu, Ilwoo Lyu, "SPHARM-Reg: Unsupervised Cortical Surface Registration using Spherical Harmonics," *IEEE Transcation on Medical Imaging*, 2024.

[C1] **Seungeun Lee**, "Facial Texure Perceiver: Towards High-Fidelity Facial Texture Recovery with Input-Level Inductive Biased Perceiver IO," *ICASSP*, 2023.

HONOR & AWARD

2024. **3rd Place**, *2nd RHOBIN Challenge at CVPR 2024.*
Sponsor: Apple and Meshcapade
Task: 3D human contact estimation from single-view images.

2020. **1st Place**, *I-GPS: Inha Group for Problem Solving.*
Institute: Inha University
Task: Developed a technology that diagnoses users' depression by analyzing structured data collected from smartphone app sensors using machine learning algorithms.

SKILL

- *Main Language & Deep Learning Library*: Python and Pytorch
- *Tools*: FreeSurfer, Blender
- *Research Domain*: Geometric Deep Learning on the Sphere, 3D Shape/Surface Analysis and Cortical Surface Registration

ACADEMIC ACTIVITY

Reviewer: MICCAI (2025∼), ICML (2026∼), TMI(2026∼), NeurIPS(2026∼)

TEACHING EXPERIENCE

Teaching Assistant: 3D Medical Image Processing and Analysis, UNIST, 2022.
Introduction to Deep Generative Models, LG Electronics, 2021.
Introduction to AI Programming, UNIST, 2021.